

Two positive regimes for uniform IDS approximation

Algebraic local coefficients and arbitrary finite-spectrum limits in arXiv:1304.0945

Automated mathematics run `fa_banach_001`, lane 7

9 August 2026

Status. Candidate partial results; current verdict: *likely valid*. Human review is recommended. The packet proves uniform IDS approximation (i) for every fixed finite set of real algebraic local coefficients and (ii) for arbitrary complex local coefficients whenever the limiting IDS has finite support. It does not settle the remaining infinite-spectrum transcendental case.

1 Source question

The source is Felix Pogorzelski, *Convergence theorems for graph sequences*, arXiv:1304.0945, Section 6.2, page 12 [4]. For a weakly convergent bounded-degree graph sequence and a fixed finite-range pattern-invariant selfadjoint operator sequence, the source recalls pointwise convergence of spectral atom masses for integer-valued local kernels (Theorem 6.10). It then asks whether the resulting pointwise, equivalently uniform, convergence remains true for arbitrary coefficients.

Transcription. After deriving uniform convergence for bounded integer-coefficient operators, the source states: “To the knowledge of the author, this is still an open question for arbitrary coefficients.”

2 Set-up

Let (G_n) be finite graphs of maximum degree at most d , with $|V(G_n)| \rightarrow \infty$, which converge weakly (in the Benjamini–Schramm sense). Let H_n be selfadjoint matrices arising from one fixed pattern-invariant kernel rule of range R : every row is determined by the rooted R -ball, and the rule does not change with n . There are only finitely many rooted R -ball types and only finitely many vertices in each such ball, so the set S of all matrix-entry values used by this rule is finite.

Write

$$N_n = |V(G_n)|, \quad \nu_n = \frac{1}{N_n} \sum_{j=1}^{N_n} \delta_{\lambda_{n,j}}.$$

Weak graph convergence and locality imply that all moments of ν_n converge. Uniform boundedness of the matrices makes the supports compact, so ν_n converges weakly to the source paper’s density-of-states measure ν ; this is source Theorem 6.6 [4].

It can already be noticed in the group situation that one cannot expect the Banach space convergence of Theorem 4.2 to hold for non-amenable structures. Indeed, it has been shown in [12] (Proposition 4.1) that sofic approximations of groups are hyperfinite if and only if the group under consideration is amenable. Thus, we infer from Proposition 2.8 that our notion of almost-additive functions will not allow for normalized convergence along approximating sequences in *non-amenable* groups. However, as far as the question of pointwise (or uniform) convergence of spectral distribution functions is concerned, one can bypass this problem by extending the algebraic eigenvalue conjecture (see e.g. [5]) to the general geometric case of graph sequences. This issue has been addressed by ABÉRT, THOM and VIRÁG in [2]. The method is to use the number theoretic techniques developed in [31] for the proof of the conjecture in the case of sofic groups. Precisely, if ν_n and ν are the positive measures associated with N_{H_n} and $N_{[\mathcal{H}]}$ respectively, then the following assertion holds true.

Theorem 6.10 (cf. [2]). *Let $(G_n) \subseteq \mathcal{G}_d$ be a weakly convergent graph sequence. Assume that $[\mathcal{H}] := (H_n)$ is an associated weakly convergent operator sequence with integer coefficients (admissible kernel functions), i.e. for all $\alpha \in \mathcal{A}_d$, we have that h_α takes its values in $\mathbb{Z}$. Then,*

$$\lim_{n \rightarrow \infty} \nu_n(\{\lambda\}) = \nu(\{\lambda\}) \quad (6.1)$$

for every $\lambda \in \mathbb{R}$.

Now well-known arguments can be used to derive from the limit relation (6.1) the uniform convergence of (N_{H_n}) , see e.g. [21], Lemma 6.3. Thus, one obtains the uniform existence of the integrated density of states for bounded, integer coefficient operators on all approximable, discrete structures with vertex degree bound. To the knowledge of the author, this is still an open question for arbitrary coefficients, see the remark following Theorem 4.4 in [31].

These elaborations show that strong spectral approximation results hold true even if one does not have a Banach space valued convergence theorem at hand. However, one may raise the question whether one can find a class of Banach space valued functions which comprises spectral distributions

Figure 1: Theorem 6.10 and the arbitrary-coefficient question on page 12 of the source paper.

3 Idea of the proof

For integer matrices, the Abért–Thom–Virág argument uses the fact that a nonzero product of eigenvalue differences is a nonzero integer. Hence it cannot be too small. With algebraic entries, that product need not be a rational integer. The replacement is to put all coefficients in one number field K and take the field norm: the product is a nonzero algebraic integer, so its norm down to $\mathbb{Q}$ is a nonzero rational integer.

The other embeddings of K replace H_n by Galois-conjugate matrices. They need not be selfadjoint, but their entries and row sparsity are uniformly bounded, so all their eigenvalues stay in one fixed disk. The conjugate factors in the field norm therefore cost only $\exp(O(N_n^2))$. This gives exactly the scale needed to rule out a positive proportion of distinct eigenvalues collapsing into an arbitrarily small punctured interval.

A second elementary number-field observation handles large multiplicities. If an eigenvalue has normalized multiplicity at least $1/k$, its degree over K is at most k . Its absolute conjugates are uniformly bounded, so only finitely many such algebraic integers exist. Avoiding this finite set and using the discriminant estimate gives a punctured-neighborhood bound tending to zero. Portmanteau’s inequalities then force convergence at every atom.

There is also a coefficient-free mechanism when the limiting measure has finite support. If $\text{supp } \nu = \{x_1, \dots, x_m\}$, put $p(t) = \prod_j (t - x_j)$. Moment convergence makes the normalized Hilbert–

Schmidt norm of $p(H_n)$ tend to zero. Locality and the fixed finite coefficient alphabet imply that the entries of all $p(H_n)$ range over one finite set. Hence every nonzero row has a uniformly positive contribution to the Hilbert–Schmidt norm, so $\text{rank } p(H_n) = o(N_n)$. Almost the entire spectrum of H_n is therefore exactly contained in $\{x_1, \dots, x_m\}$, which forces convergence of the individual atom masses.

4 Main result

Theorem 1 (Algebraic-coefficient pointwise approximation). *In the set-up above, assume that every value in the finite local coefficient set S is a real algebraic number. Then, for every $x \in \mathbb{R}$,*

$$\lim_{n \rightarrow \infty} \nu_n(\{x\}) = \nu(\{x\}).$$

Consequently, if

$$F_n(t) = \nu_n((-\infty, t]) \quad \text{and} \quad F(t) = \nu((-\infty, t]),$$

then

$$\lim_{n \rightarrow \infty} \|F_n - F\|_\infty = 0.$$

Remark 1. The theorem strictly extends the integer hypothesis in source Theorem 6.10: it includes arbitrary real algebraic local values in a common number field. It does not include a coefficient set containing a transcendental number.

Theorem 2 (Arbitrary coefficients with finite limiting spectrum). *In the set-up above, impose no arithmetic restriction on the finite set $S \subset \mathbb{C}$. If the weak limiting measure ν has finite support, then for every $x \in \mathbb{R}$,*

$$\lim_{n \rightarrow \infty} \nu_n(\{x\}) = \nu(\{x\}),$$

and consequently $\|F_n - F\|_\infty \rightarrow 0$.

The next proposition does not prove convergence in the remaining case, but it sharply restricts any possible failure. It applies to more general sparse matrix families and uses Jaikin-Zapirain’s strict eigenvalue theorem [3, Theorem 2.2].

Proposition 1 (Macroscopic eigenvalues come from a fixed finite set). *Let A_n be square complex matrices of orders N_n . Suppose that every row and every column has at most D nonzero entries and that all entries belong to one fixed finite set $S \subset \mathbb{C}$. For every $\eta > 0$ there is a finite set $E_\eta \subset \mathbb{C}$, independent of n , such that*

$$\frac{\dim \ker(A_n - \lambda I)}{N_n} \geq \eta \quad \implies \quad \lambda \in E_\eta.$$

In particular, this holds for every fixed-range pattern-invariant operator sequence, without any algebraicity assumption.

5 Proof

Multiplying every H_n by the same positive integer only rescales the spectral variable and does not affect either conclusion. Since S is finite, we may therefore choose a finite Galois extension $K/\mathbb{Q}$, realized inside $\mathbb{C}$ with its distinguished identity embedding, such that

$$S \subset \mathcal{O}_K \cap \mathbb{R}.$$

For $\sigma \in \text{Gal}(K/\mathbb{Q})$, let H_n^σ be obtained by applying σ entrywise. There is a constant $M \geq 1$, independent of n and σ , such that every row sum of absolute values of every H_n^σ is at most M . Indeed, the number of nonzero entries per row is bounded in terms of d, R , and there are only finitely many numbers $\sigma(s)$ with $s \in S$. Thus every eigenvalue of every H_n^σ has modulus at most M .

Lemma 1 (Only finitely many possible large atoms). *For every positive integer k , there is a finite set X_k of algebraic integers, depending only on K, M, k , such that*

$$\nu_n(\{\alpha\}) \geq \frac{1}{k} \implies \alpha \in X_k$$

for every n and every eigenvalue α of H_n .

Proof. Let α have multiplicity m as a root of $p_n(t) = \det(tI - H_n) \in \mathcal{O}_K[t]$. If q_α is its minimal polynomial over K , separability in characteristic zero gives $q_\alpha^m \mid p_n$. Therefore

$$m[K(\alpha) : K] \leq N_n.$$

If $m/N_n \geq 1/k$, then $[K(\alpha) : K] \leq k$. Moreover α is integral over $\mathcal{O}_K$, hence is an algebraic integer over $\mathbb{Z}$.

Every embedding of $\mathbb{Q}(\alpha)$ into $\mathbb{C}$ extends to an embedding of $K(\alpha)$. Its restriction to K is some σ , and the image of α is an eigenvalue of H_n^σ ; hence every absolute conjugate of α has modulus at most M . Finally,

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] \leq [K : \mathbb{Q}] k.$$

There are only finitely many algebraic integers of bounded degree whose conjugates all have bounded modulus: their monic minimal polynomials have bounded degree and bounded integer coefficients. Taking all such numbers as X_k proves the claim. $\square$

Lemma 2 (Number-field anti-clustering estimate). *Put $q = [K : \mathbb{Q}]$. Fix $x \in \mathbb{R}$ and $k \geq 1$. Choose $0 < \varepsilon < 1/2$ such that*

$$I_{x,\varepsilon} := (x - \varepsilon, x + \varepsilon) \setminus \{x\} \quad \text{satisfies} \quad I_{x,\varepsilon} \cap X_k = \emptyset.$$

Then, for every n ,

$$\nu_n(I_{x,\varepsilon}) \leq \left(\frac{q \log(2M)}{\log(1/(2\varepsilon))} + \frac{1}{k} \right)^{1/2}. \quad (1)$$

Proof. List the eigenvalues of H_n , with multiplicity, as $\lambda_1, \dots, \lambda_N$, where $N = N_n$. Define the nonzero-pair discriminant

$$D_n = \prod_{\substack{1 \leq i < j \leq N \\ \lambda_i \neq \lambda_j}} (\lambda_i - \lambda_j)^2.$$

It is a nonzero algebraic integer. It is also fixed by every automorphism of $\overline{\mathbb{Q}}$ fixing K : such an automorphism permutes the multiset of roots of p_n , preserves equality and inequality of roots, and the square removes the sign. Thus

$$D_n \in \mathcal{O}_K \setminus \{0\}.$$

Its field norm is a nonzero rational integer. For every nonidentity $\sigma \in \text{Gal}(K/\mathbb{Q})$, the conjugate $\sigma(D_n)$ is the analogous nonzero-pair product for H_n^σ . Since all conjugate eigenvalues have modulus at most M ,

$$1 \leq |N_{K/\mathbb{Q}}(D_n)| \leq |D_n| (2M)^{(q-1)N^2}.$$

Consequently,

$$|D_n| \geq (2M)^{-(q-1)N^2}. \quad (2)$$

Set

$$a = \nu_n(I_{x,\varepsilon}), \quad b = \nu_n(\{x\}), \quad s = \sum_{\lambda \in I_{x,\varepsilon}} \nu_n(\{\lambda\})^2.$$

No eigenvalue in $I_{x,\varepsilon}$ has mass at least $1/k$, by Lemma 1; hence

$$s < \frac{1}{k}. \quad (3)$$

The normalized number of *ordered* pairs of unequal eigenvalue occurrences which both lie in $(x - \varepsilon, x + \varepsilon)$ is exactly

$$(a + b)^2 - (b^2 + s) = a^2 + 2ab - s.$$

The square in D_n makes its factors correspond precisely to ordered unequal pairs. Factors belonging to the displayed collection have modulus at most 2ε , while all remaining factors have modulus at most $2M$. Therefore

$$|D_n| \leq (2\varepsilon)^{N^2(a^2+2ab-s)} (2M)^{N^2}. \quad (4)$$

Combining (2) and (4) gives

$$a^2 + 2ab - s \leq \frac{q \log(2M)}{\log(1/(2\varepsilon))}.$$

Since $b \geq 0$, estimate (3) now yields (1). □

Proof of Theorem 1. Fix $x \in \mathbb{R}$. For each k , Lemma 1 allows us to choose

$$0 < \varepsilon_k < \min\{1/k, 1/2\}, \quad I_{x,\varepsilon_k} \cap X_k = \emptyset.$$

Let

$$\delta_k = \left(\frac{q \log(2M)}{\log(1/(2\varepsilon_k))} + \frac{1}{k} \right)^{1/2}.$$

Then $\delta_k \rightarrow 0$ and Lemma 2 says

$$\sup_n \nu_n(I_{x,\varepsilon_k}) \leq \delta_k. \quad (5)$$

Weak convergence and the Portmanteau theorem give

$$\limsup_{n \rightarrow \infty} \nu_n(\{x\}) \leq \nu(\{x\}). \quad (6)$$

On the other hand, for the open interval $J_k = (x - \varepsilon_k, x + \varepsilon_k)$,

$$\begin{aligned} \nu(\{x\}) &\leq \nu(J_k) \leq \liminf_{n \rightarrow \infty} \nu_n(J_k) \\ &\leq \liminf_{n \rightarrow \infty} \nu_n(\{x\}) + \delta_k, \end{aligned}$$

where the last inequality follows from (5) by taking a subsequence realizing the lower limit of the singleton masses. Letting $k \rightarrow \infty$ and combining with (6) proves

$$\nu_n(\{x\}) \longrightarrow \nu(\{x\}).$$

Finally, the standard distribution-function lemma says that weak convergence of probability measures together with convergence of the mass of every singleton is equivalent to uniform convergence of their cumulative distribution functions. For completeness, one proves it by isolating the

finitely many atoms of ν larger than a prescribed tolerance, choosing a finite partition by continuity points of ν on the remaining intervals, and using monotonicity between consecutive partition points. The atom convergence controls the isolated jumps. Applying the lemma to $\nu_n \Rightarrow \nu$ completes the proof. $\square$

Proof of Theorem 2. Write $\text{supp } \nu = \{x_1, \dots, x_m\}$ and set

$$p(t) = \prod_{j=1}^m (t - x_j).$$

The spectra of all H_n lie in one fixed compact interval. Weak convergence therefore gives

$$\frac{1}{N_n} \|p(H_n)\|_{\text{HS}}^2 = \frac{1}{N_n} \text{Tr}(p(H_n)^2) = \int_{\mathbb{R}} |p(t)|^2 d\nu_n(t) \longrightarrow \int_{\mathbb{R}} |p(t)|^2 d\nu(t) = 0. \quad (6)$$

The matrices $p(H_n)$ are again fixed-range pattern-invariant matrices: their range is at most mR , and an entry is determined by a rooted ball of a fixed radius together with two marked vertices. Bounded degree leaves only finitely many such marked ball types. Because the original local rule uses only the fixed finite alphabet S , all entries of all the matrices $p(H_n)$ therefore belong to one finite set $T \subset \mathbb{C}$. If $T \setminus \{0\}$ is nonempty, let

$$c = \min\{|t| : t \in T \setminus \{0\}\} > 0;$$

if it is empty, the conclusion below is immediate. Every nonzero row of $p(H_n)$ contributes at least c^2 to its squared Hilbert–Schmidt norm. Thus (6) implies

$$\text{rank } p(H_n) \leq \#\{\text{nonzero rows of } p(H_n)\} \leq c^{-2} \|p(H_n)\|_{\text{HS}}^2 = o(N_n). \quad (7)$$

Since H_n is selfadjoint, functional calculus and (7) give

$$\sum_{j=1}^m \nu_n(\{x_j\}) = \frac{\dim \ker p(H_n)}{N_n} \longrightarrow 1. \quad (8)$$

Portmanteau gives $\limsup_n \nu_n(\{x_j\}) \leq \nu(\{x_j\})$ for each j , while the right-hand atom masses sum to one. Equation (8) forces equality in each of these finitely many upper bounds, hence $\nu_n(\{x_j\}) \rightarrow \nu(\{x_j\})$. If x is not among the x_j , Portmanteau gives $\nu_n(\{x\}) \rightarrow 0 = \nu(\{x\})$. The distribution-function lemma used above now yields uniform convergence. $\square$

Proof of Proposition 1. Regard the support of A_n as a bipartite graph between a row copy and a column copy of $\{1, \dots, N_n\}$, keeping the matrix value as an edge label. For each nonzero $s \in S$, König’s edge-colouring theorem partitions the s -labelled edges into at most D matchings. Consequently there are a fixed number $r \leq D|S \setminus \{0\}|$ and partial permutation matrices $P_{n,1}, \dots, P_{n,r}$ such that

$$A_n = \sum_{i=1}^r c_i P_{n,i}, \quad (9)$$

where the coefficient list (c_i) is independent of n (zero matchings may be inserted when needed).

Every partial permutation matrix P is the average of two signed permutation matrices. Indeed, extend its bijection from its domain to its range by an arbitrary bijection between the two complements. Give the extension the same positive signs on the original domain in two copies, and

opposite signs on the complementary domain. If the resulting signed permutation matrices are U and V , then $P = (U + V)/2$.

Let F be the free group on $2r$ fixed generators and let

$$a = \sum_{i=1}^r \frac{c_i}{2} (u_i + v_i) \in \mathbb{C}[F].$$

For each n , mapping u_i, v_i to the signed permutations just constructed defines a representation ρ_n of F on $\mathbb{C}^{N_n}$ with $\rho_n(a) = A_n$.

Replace every signed permutation action on $\{1, \dots, N_n\}$ by its standard double cover on $X_n = \{1, \dots, N_n\} \times \{+1, -1\}$: a signed permutation sends (j, ϵ) to $(\sigma(j), s_j \epsilon)$. This is an ordinary finite F -set. The odd subspace of $\mathbb{C}[X_n]$ is invariant and its representation is unitarily equivalent to ρ_n . Hence an eigenvalue of A_n of multiplicity at least ηN_n is an eigenvalue of the permutation operator associated with the fixed group-ring element a on X_n , with multiplicity at least $(\eta/2)|X_n|$. Jaikin-Zapirain's strict eigenvalue theorem says that the union of all eigenvalues with such normalized multiplicity, over all finite F -sets, is finite. Taking that finite set proves the claim. $\square$

Corollary 1 (Necessary form of a counterexample). *If atom convergence fails at x for an arbitrary-coefficient local operator sequence, then along a subsequence a positive amount of spectral mass lies in every punctured neighbourhood of x , while the normalized multiplicity of each eigenvalue in a sufficiently small such neighbourhood is arbitrarily small. Thus failure can only occur through a diffuse cloud of distinct eigenvalues coalescing at x , not through a moving macroscopic atom. By Theorem 2, the limiting IDS of any counterexample must have infinite support.*

Proof. Failure and Portmanteau imply that, along a subsequence, there is a $\delta > 0$ such that $\nu_n(\{x\}) \leq \nu(\{x\}) - \delta$. For every open interval J containing x , Portmanteau also gives $\liminf_n \nu_n(J) \geq \nu(J) \geq \nu(\{x\})$; hence eventually $\nu_n(J \setminus \{x\}) \geq \delta/2$ along that subsequence. Given $\eta > 0$, Proposition 1 supplies a finite E_η . Choose J so small that $(J \setminus \{x\}) \cap E_\eta = \emptyset$. Every eigenvalue contributing to the displayed punctured mass then has normalized multiplicity less than η . The last assertion is the contrapositive of Theorem 2. $\square$

6 Verification notes and limitations

Proof audit. The argument was checked independently at the following structural points: (i) one denominator clears the finite coefficient set; (ii) multiplicity bounds relative degree over K ; (iii) the squared nonzero-pair discriminant belongs to $\mathcal{O}_K$ despite repeated eigenvalues; (iv) the field norm and the row-sum bounds on every conjugate matrix give (2); (v) the Portmanteau inequalities have the required directions; (vi) the finite-spectrum proof uses a genuinely finite entry alphabet for $p(H_n)$; and (vii) the signed-permutation double cover puts the macroscopic-eigenvalue question exactly within the scope of the strict eigenvalue theorem. No numerical experiment is used as proof evidence.

Human-review focus. The most important lines to verify are $D_n \in \mathcal{O}_K$ in Lemma 2, the assertion that the entries of $p(H_n)$ range over one finite set in Theorem 2, and the reduction from partial permutations to a fixed free-group element in Proposition 1.

Scope. The original arbitrary-real-coefficient problem remains open when the limit has infinite support. The algebraic proof uses algebraicity exactly through the nonzero field-norm lower bound and the finiteness of bounded-house algebraic integers of bounded degree. Neither input is available for a general transcendental coefficient tuple. The coefficient-free rank proof requires one polynomial to vanish on the whole limiting support and hence does not extend directly beyond finite support. Proposition 1 excludes only a moving eigenvalue with one macroscopic eigenspace; it does not exclude diffuse coalescence of increasingly many small eigenspaces.

7 Novelty check

The exact graph-sequence formulation was not located in a bounded search. The search covered the local run indexes; the phrases *algebraic coefficients Lück approximation spectral measures graph sequences*, *number field Lück approximation*, and *arbitrary coefficients graph sequences spectral distribution*; and arXiv:1304.0945, arXiv:math/0701294, the Abért–Thom–Virág preprint cited in the source, and arXiv:2603.01610.

The number-field mechanism is not itself new: Thom proves the algebraic coefficient analogue for sofic group approximations [5]. Jaikin-Zapirain proves arbitrary-coefficient Lück approximation for fixed group-ring operators along deterministic sofic approximations and, separately, the strict eigenvalue theorem used in Proposition 1 [3]. The 2026 work of Donoso-Echenique, Pogorzelski, and Schrödl-Baumann proves pointwise convergence for finitely many rational values in a random-operator/sofic-group setting and treats arbitrary complex coefficients through adapted approximants with varying rational coefficients [2]. That paper explicitly says that arbitrary fixed real or complex values remain open in its random setting and asks whether Jaikin-Zapirain’s methods extend there. No searched source stated Theorem 1, Theorem 2, or Corollary 1 for every weakly convergent bounded-degree graph sequence. Accordingly, novelty of these extensions is *provisional*, not established.

References

- [1] M. Abért, A. Thom, and B. Virág, *Benjamini–Schramm convergence and pointwise convergence of the spectral measure*, circulated preprint, 2013.
- [2] M. Donoso-Echenique, F. Pogorzelski, and M. Schrödl-Baumann, *Random operators, spectral measures, and local empirical convergence in sofic groups*, arXiv:2603.01610, 2026.
- [3] A. Jaikin-Zapirain, *The base change in the Atiyah and the Lück approximation conjectures*, Geometric and Functional Analysis 29 (2019), 464–538.
- [4] F. Pogorzelski, *Convergence theorems for graph sequences*, International Journal of Algebra and Computation 24 (2014), 1233–1251; arXiv:1304.0945.
- [5] A. Thom, *Sofic groups and diophantine approximation*, Communications on Pure and Applied Mathematics 61 (2008), 1155–1171; arXiv:math/0701294.